Grounded Confidence with Selective LLM Auditing for Reliable Long-Tailed Radiology Reports

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Abstract. Reliable disease-specific confidence is critical for clinically trustworthy radiology report generation (RRG). Recent RRG systems integrate constructed information from multi-label classifiers into LLM prompting pipelines, yet existing methods do not mitigate the calibration of the predictive confidence which is crucial in medical domain. Moreover, long-tailed clinical distributions and Tail-Compensatory Learning (TCL) strategies often induce systemic overconfidence in rare diseases. To address this, we propose a unified two-stage calibration framework for trustworthy RRG. First, Adaptive Confidence Distillation (ACD) performs statistical calibration by distilling adaptively balancing temperature scaled distribution to mitigate long-tail bias while preserving discriminative performance. Second, a Grounded Meta-Calibrator (GMC) verifies predictions against image-grounded anatomical and clinical evidence, with a Selective LLM Auditing pipeline adjudicating final confidence. Experiments on MIMIC-CXR under both positive-oversampling and stratified sampling settings demonstrate that our approach improves classification accuracy, calibration metrics, and report reliability, while effectively reducing overconfidence in tail-class diseases.

Keywords: Radiology Report Generation · Long-Tailed Calibration · LLM Auditing.

1 Introduction

Automated Radiology Report Generation (RRG) [6, 3, 13] has emerged as a promising tool for chest X-ray interpretation, providing decision support to clinicians. Recent advances have significantly improve report quality by incorporating structured diagnostic metadata from multi-label classifiers into Large Language Model (LLM) prompts [7, 8]. However, the clinical integration of RRG systems fundamentally relies on their reliability—specifically, the calibration of their predictive confidence. In safety-critical settings, predicted probabilities must accurately reflect true correctness, as overconfident errors can propagate to LLMs and produce confidently hallucinated diagnoses.

Prior work on improving RRG reliability, including logit adjustment [17, 20, 22] and visually grounded methods [2, 23, 11], mainly addresses class imbalance

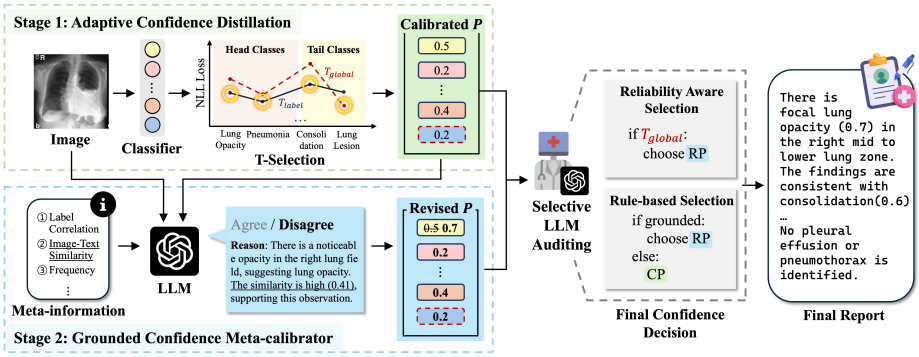

Fig. 1: Overall framework of the proposed dual-stage calibration method. P is probability of each label, CP is Calibrated P from Stage 1, and RP is Revised P from Stage 2.

and spatial interpretability, but does not provide well-calibrated disease-level confidence scores. While Tail-Compensatory Learning (TCL) [4, 14, 21] is often used for long-tailed clinical data, we observe a critical trade-off in TCL: distortion of disease prevalence, causing systemic overconfidence in tail classes. When propagated to the LLM, this miscalibration compromises report trustworthiness. To uncover the root cause of this systemic overconfidence, we first analyze the calibration-accuracy trade-off induced by TCL under long-tailed clinical distributions. This analysis reveals that while TCL enhances rare disease recall, it fundamentally distorts the underlying logit distribution—an issue that standard post-hoc calibration methods fail to fully resolve.

Motivated by these findings, we propose a unified two-stage calibration strategy that seamlessly integrates statistical calibration with knowledge-based grounded verification to correct confidence distribution-level distortion and validate instance-level predictions. Specifically, our framework consists of two complementary stages. First, **Adaptive Confidence Distillation (ACD)** performs stable statistical calibration during training. It adaptively balances label-specific and global temperature scaling based on Negative Log-Likelihood (NLL) [10] values, effectively mitigating long-tail bias while preserving discriminative performance. Second, to address cases where calibrated probabilities may still lack factual grounding, we introduce a **Grounded Meta-Calibrator (GMC)**. GMC validates the initial predictions against image-grounded anatomical evidence and clinical metadata. Finally, a Selective LLM Auditing pipeline adjudicates the final confidence by weighing the stability of the ACD-based scaling against the explicit rule-based evidence strength from the GMC.

The main contributions of this work are summarized as follows:

- We provide an in-depth analysis of the impact of TCL, revealing the significant trade-off between tail-class performance improvements and increased calibration errors under long-tailed clinical distributions.

- We propose a novel, trustworthy RRG framework that unifies ACD for robust statistical calibration and a GMC for knowledge-based verification, finalized through Selective LLM Auditing pipeline.
- Extensive evaluations on the MIMIC-CXR [18] dataset under both positive-oversampling and stratified settings demonstrate that our approach substantially improves classification accuracy, calibration metrics, and report reliability, effectively curbing tail-class overconfidence.

2 Method

2.1 Analysis on Calibration for Long-Tailed Radiology Reports

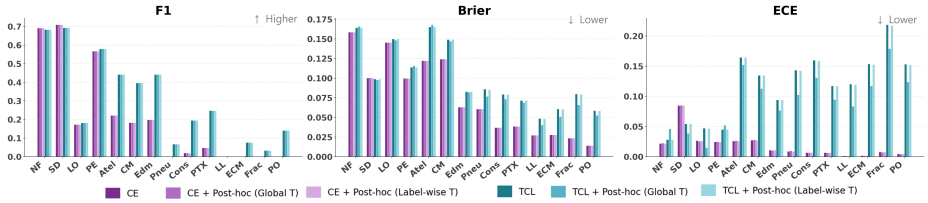

Fig. 2: Comparison between CE and TCL illustrating the trade-off between tail-class performance and calibration. The progressively lighter shading bars represent uncalibrated, post-hoc global τ scaled and label-wise τ scaled, respectively.

To explicitly demonstrate the trade-off between predictive accuracy and confidence calibration, we compare four learning and calibration settings across three key metrics: F1 score (for classification performance considering class imbalance), Expected Calibration Error (ECE) [12], and Brier score (for calibration reliability) [9]. We first compare standard Cross-Entropy (CE) learning with TCL. TCL emphasizes rare diseases by assigning larger losses to positive labels and oversampling rare cases. As shown in Fig. 2, while TCL significantly improves the F1 score on tail classes over CE, it severely distorts the overall logit distribution, inducing systemic overconfidence and drastically worsening both ECE and Brier scores. To investigate whether standard post-hoc calibration can resolve this distortion, we apply Temperature Scaling (TS) to the TCL models. TS re-scales logits using a temperature parameter τ optimized on a validation set:

$$p_i^{\mathrm{TS}} = \sigma\left(\frac{z_i}{\tau}\right), \tag{1}$$

where z_i denotes the logit of class i, and $\sigma(\cdot)$ is the sigmoid function. We evaluate two TS variants based on different τ estimation strategies: global and label-wise. Let $\mathrm{NLL}(\tau)$ denote the validation NLL. For each label $\ell \in \{1, \ldots, L\}$, we compute the optimal label-wise $\tau_\ell^{\mathrm{Label}}$, and define the global τ^{Global} as their

mean:

$$\tau_\ell^{\text{Label}} = \arg\min_{\tau>0} \text{NLL}_\ell(\tau), \quad \tau^{\text{Global}} = \frac{1}{L}\sum_{\ell=1}^{L} \tau_\ell^{\text{Label}} \tag{2}$$

Applying TS with τ^{Global} consistently reduces ECE and Brier scores across classes. However, this improvement is largely uniform, reflecting overall confidence shrinkage rather than structural correction of the tail-class distortion. Conversely, applying TS with τ_ℓ^{Label} provides limited additional benefit, particularly for tail classes where calibration errors remain substantial. This failure occurs because the scarcity of validation samples for rare diseases leads to highly unstable and high-variance τ estimation. These observations confirm that post-hoc scaling cannot fundamentally correct the distortion introduced by TCL.

2.2 Adaptive Confidence Distillation

Building upon the above analysis, we propose ACD, which corrects class-conditional miscalibration during training rather than via post-hoc adjustment. Instead of relying purely on post-hoc TS, ACD incorporates τ-scaled targets into training via a KL-divergence [19] objective. For tail classes, τ_ℓ^{Label} may be unreliable due to high NLL variance from limited validation samples. To avoid unstable calibration targets, such labels adopt the more stable global τ, producing smoother target distributions. For reliable labels, we retain the label-wise τ to preserve class-specific calibration. Formally, for each label ℓ, we select the τ yielding the lower validation NLL:

$$\tau_\ell^{\text{ACD}} = \begin{cases} \tau_\ell^{\text{Label}}, & \text{if } \text{NLL}_\ell(\tau_\ell^{\text{Label}}) \le \text{NLL}_\ell(\tau^{\text{Global}}), \\ \tau^{\text{Global}}, & \text{otherwise.} \end{cases} \tag{3}$$

Finally, ACD minimizes the KL divergence between the selected τ scaled validation distribution and the original training distribution:

$$\mathcal{L}_{\text{ACD}} = \sum_{\ell=1}^{L} \text{KL}\big(p_\ell^{\text{val}}(z/\tau_\ell^{\text{ACD}}) \parallel p_\ell^{\text{train}}(z)\big)\,. \tag{4}$$

To establish robust discrimination before calibration, we first train the model for 5 epochs with TCL and CE loss, then introduce $\mathcal{L}_{\text{ACD}}$ with a gradually increasing weight λ. During this stage, the validation-based τ is re-estimated every two epochs and used to update the τ-scaled teacher distribution. The total objective is defined as:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{ACD}} \tag{5}$$

We observe that the number of labels selecting τ_ℓ^{Label} increases from 3 to 5 and eventually to 12 during training, indicating increasingly reliable label-wise τ estimation and more stable calibration.

2.3 Grounded Meta-Calibrator and Selective LLM Auditing

In the earlier stage, ACD calibrates probabilities from the perspective of multi-label classification. However, due to limited training samples for rare diseases, the classifier's predicted label presence and the associated probability that we interpret as confidence can still be incorrect.

Therefore, we propose the GMC as a complementary calibration module. GMC leverages meta-information from training data, including label frequency, label correlation, and image–label similarity computed using a medical CLIP model [29]. Based on this information, it determines whether it agrees or disagrees with the ACD-derived confidence. When it disagrees, it generates a grounded reason from provided meta-information and outputs a revised probability. The final decision is determined by reliability aware selection based on the selected τ type for each label. If a global τ is used, the ACD probability is considered less reliable and the *GMC-Revised Probability*(RP) is adopted. If a label-wise τ is selected, the RP is used only when it is strongly grounded; otherwise, the original *ACD-Calibrated Probability*(CP) is retained. Grounding is verified via rule-based keyword matching over 75 anatomical and visual categories with negation-aware filtering to exclude absent findings.

This mechanism ensures that well-calibrated ACD probabilities are overridden only when supported by explicit clinical evidence. The final confidence probabilities are included in the prompt to LLM report generator, and displayed alongside each label to provide intuitive confidence cues in final report.

3 Experiment and Results

3.1 Experimental Setup

We evaluate our method on a MIMIC-CXR [18] subset of 75,216 image-report pairs from 12,967 patients. Data are split at the patient level into train/val/test (8:1:1) to prevent leakage. GT labels for 14 categories are extracted via CheXpert [16]. For robust RRG evaluation, we construct two additional 500-sample test sets using positive-oversampling and stratified sampling. For ACD, an ImageNet-pretrained ResNet-50 is fine-tuned with Adam ($lr = 1 \times 10^{-3}$) for 10 epochs (5 with CE, followed by 5 with distillation and a 3-epoch linear ramp-up to $\lambda = 1$), updating temperature every two epochs. GPT-4o is used for two sequential inference passes per sample. All experiments are conducted on RTX 3090 GPUs.

3.2 Main Results

Multi-Label Classification-Calibration Trade-off Analysis We evaluate performance from two complementary perspectives: classification accuracy and calibration reliability. For classification, we report AP and F1 score. For calibration, we report the Brier score [9] for squared prediction error and ECE [12] for confidence–accuracy gap. As shown in Table 1, CE performs well on overall classes, but struggles on tail classes. TCL improves the F1 on tail classes

Table 1: Multi-label classification (Tail in parentheses). **Bold**: best, underline: second-best. (F: Focal, B: Brier, PL/PG: Post-hoc Label-wise/Global τ.)

	Classification (↑)		Calibration (↓)	
Strategy	AP	F1 Score	Brier	ECE
CE	0.310 (0.080)	0.206 (0.007)	0.074 (0.026)	0.014 (0.007)
TCL	0.296 (0.077)	0.299 (0.100)	0.101 (0.064)	0.116 (0.153)
TCL_F	**0.303** (**0.082**)	0.296 (0.080)	0.100 (0.055)	0.125 (0.132)
TCL_B	0.288 (0.067)	0.258 (0.036)	**0.088** (**0.036**)	**0.062** (**0.053**)
TCL_PL	0.296 (0.077)	0.299 (0.100)	0.101 (0.064)	0.115 (0.152)
TCL_PG	0.296 (0.077)	0.299 (0.100)	0.097 (0.056)	0.090 (0.120)
TCL_ACD	0.303 (0.080)	**0.318** (**0.101**)	0.101 (0.055)	0.120 (0.119)

Table 2: RRG generation performance under different sampling strategies.

	Oversampled (↑)			Stratified (↑)		
Strategy	BLEU	BERTScore	GREEN	BLEU	BERTScore	GREEN
Image-only	0.129	0.313	0.246	0.115	0.286	0.206
Image+Meta	0.125	0.311	0.212	0.104	0.286	0.175
Image+CLS	0.128	0.312	0.268	0.108	0.287	0.239
PromptMRG [17]	**0.163**	0.296	**0.299**	**0.142**	0.275	0.217
MAIRA-2 [2]	0.069	0.247	0.103	0.072	0.244	0.208
Image+ACD	0.132	**0.316**	0.259	0.112	**0.292**	0.223
Image+ACD+GMC	0.124	0.309	0.282	0.106	0.287	**0.241**

but severely degrades calibration, raising ECE from 0.014 to 0.116. Focal and Brier losses reduce calibration error relative to TCL but lower F1, especially on tail classes, suggesting weakened decision boundaries. Post-hoc scaling decreases calibration error without improving classification, as it only rescales logits; PG outperforming PL further indicates suboptimal label-wise optimization. In contrast, ACD improves both overall and tail-class F1 (+0.019, +0.001) over TCL while maintaining competitive calibration. Although not always achieving the lowest ECE, ACD preserves tail-class discrimination without sacrificing classification performance, yielding a more favorable trade-off in long-tailed settings.

RRG Accuracy and Calibration Comparison We evaluate RRG performance using BLEU [25] (2-gram), BERTScore [30], and GREEN [24] which captures clinically significant errors and aligns closely with expert judgment. To evaluate calibration in RRG, we adapt the Brier score and ECE to verbalized confidence, denoted as vBrier and vECE [27, 28]. Explicit probabilities in the generated report are used as confidence scores when available; otherwise, confidence is inferred via hedging-phrase parsing. Both are evaluated against the verbalized confidence extracted from the ground-truth (GT) report. We further introduce High-Confidence Error (HCE), defined over the top 20% of predic-

Table 3: RRG calibration performance under different sampling strategies. An asterisk (*) denotes strategies without explicit predictive probabilities.

	Oversampled (↓)			Stratified (↓)		
Strategy	vECE	vBrier	HCE	vECE	vBrier	HCE
Image-only*	**0.310**	0.274	0.699	0.338	0.303	0.736
Image+Meta*	0.319	0.288	0.714	**0.333**	0.302	0.747
Image+CLS	0.362	0.275	0.490	0.364	0.278	0.468
PromptMRG* [17]	0.438	0.394	0.743	0.481	0.435	0.768
MAIRA-2* [2]	0.391	0.350	0.664	0.421	0.376	0.668
Image+ACD	0.348	**0.261**	**0.464**	0.345	**0.262**	0.396
Image+ACD+GMC	0.352	0.264	0.467	0.348	0.263	**0.394**

tions ranked by confidence magnitude $|P - 0.5|$, to quantify incorrect outputs expressed with high confidence. We refer to the baseline as Image-only, and its variants as Meta, CLS, ACD, and ACD+GMC.

In generation (Table 2), ACD slightly improves BLEU and BERTScore while maintaining competitive GREEN performance. ACD+GMC further improves GREEN, suggesting stronger clinical grounding beyond surface similarity, though lexical metrics are not always maximized. In calibration (Table 3), ACD attains the lowest or second-lowest errors in several metrics, reducing calibration gaps relative to CLS (average vECE -0.0167). ACD+GMC maintains comparable calibration performance, while refining semantic reliability. These results indicate that ACD improves distribution-level calibration, and GMC provides complementary instance-level correction.

3.3 Ablation Studies

Label-wise Effect of GMC with and without ACD Figure 3 illustrates the label-level impact of GMC and Selective LLM Auditing. On the left, GMC (orange) mainly boosts head-label positives while staying conservative on tail classes, reflecting prior-aware re-scaling. The final outputs (green) fall between the classifier and GMC, as the auditor keeps visually supported revisions and rejects over-corrections, a pattern more evident without ACD due to noisier probabilities. On the right, GREEN improvements align with Hall_{kw} reductions, suggesting that grounded corrections mitigate hallucinations. Overall, these results explain why adding GMC does not monotonically improve all metrics—while GMC sharpens clinically grounded labels via instance-level correction, its gains depend on ACD and the auditor's ability to verify evidence.

RRG with Different LLMs We evaluated our framework across four LLM APIs to assess its generalizability, as shown in Table 4. While AP is consistently higher under the ACD setting, adding GMC further improves GREEN scores for frontier models like GPT-4o [15] (reaching 0.282) and Gemini-2.5-Pro [5]. The

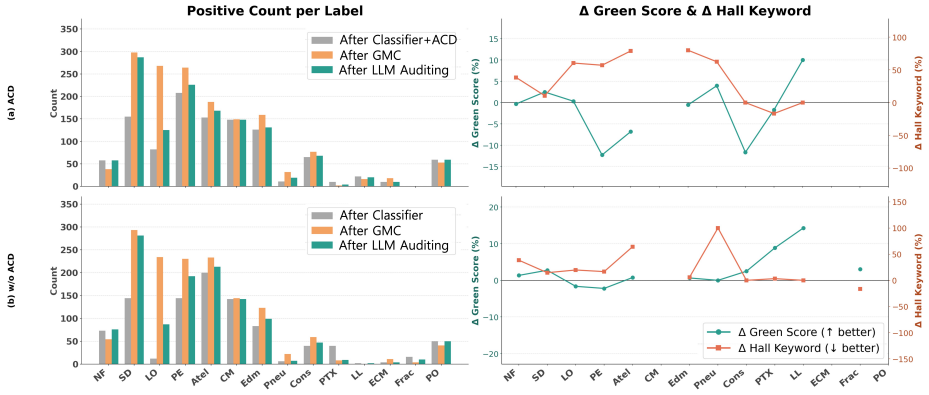

Fig. 3: Label-wise effect of GMC and Selective LLM Auditing. Left: Positive label counts sorted by frequency. Right: Relative changes in GREEN score and hallucination rates (Hall_{kw}).

Table 4: Comparison of RRG performance across four LLM APIs.

	Image+ACD		Image+ACD+GMC	
Model	AP	GREEN	AP	GREEN
GPT-4o [15]	0.334	0.259	0.300	**0.282**
Claude-4.5-Sonnet [1]	0.329	**0.281**	0.315	0.277
Gemini-2.5-Pro [5]	**0.346**	0.244	**0.324**	0.260
MedGemma [26]	0.275	0.243	0.255	0.228

increased variability in GREEN scores under the GMC setting (std: 0.025 vs. 0.018) suggests that LLMs exhibit varying sensitivities to grounded reasoning signals. While GPT-4o shows superior adjudication capabilities, MedGemma [26] performs comparatively lower, likely due to its smaller scale and limited adherence to complex calibration instructions.

4 Discussion and Conclusion

In this work, we addressed the calibration trade-off introduced by TCL under long-tailed clinical distributions. We proposed a trustworthy RRG framework that combines ACD for stable statistical calibration with GMC for knowledge-based verification, unified through Selective LLM Auditing. Experiments on MIMIC-CXR demonstrate improved calibration reliability while preserving clinically meaningful report generation, highlighting the benefit of integrating distribution-level calibration with instance-level semantic verification. Looking forward, stronger calibration objectives and improved probability-to-language alignment could further bridge statistical confidence and its verbalized expression, advancing more transparent and clinically reliable report generation.

References

1. Anthropic: Claude sonnet 4.5 system card. PDF (2025), technical system card describing model architecture and evaluation
2. Bannur, S., Bouzid, K., Castro, D.C., et al.: Maira-2: Grounded radiology report generation. arXiv (2024)
3. Bu, S., Song, Y., Li, T., Dai, Z.: Dynamic knowledge prompt for chest x-ray report generation. In: Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024). pp. 5425–5436 (2024)
4. Chamseddine, E., Mansouri, N., Soui, M., et al.: Handling class imbalance in covid-19 chest x-ray images classification: Using smote and weighted loss. Applied Soft Computing **129**, 109588 (2022)
5. Comanici, G., Bieber, E., Schaekermann, M., Pasupat, I., Sachdeva, N., Dhillon, I., Blistein, M., Ram, O., Zhang, D., Rosen, E., et al.: Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. arXiv preprint arXiv:2507.06261 (2025)
6. Delbrouck, J.B., Xu, J., Moll, J., et al.: Automated structured radiology report generation. In: Proceedings of the Association for Computational Linguistics. pp. 26813–26829 (2025)
7. Edirisinghe, D., Nimalsiri, W., Hennayake, M., et al.: Chest x-ray report generation using abnormality guided vision language model. IEEE Access (2025)
8. Fan, Y., Yang, Z., Liu, R., et al.: Medical report generation is a multi-label classification problem. arXiv preprint arXiv:2409.00250 (2024)
9. Glenn, W.B.: Verification of forecasts expressed in terms of probability. Monthly weather review **78**(1), 1–3 (1950)
10. Gonzalez-Jimenez, A., Lionetti, S., Gottfrois, P., Gröger, F., Navarini, A., Pouly, M.: Robust t-loss for medical image segmentation. Medical Image Analysis p. 103735 (2025)
11. Gu, D., Gao, Y., Zhou, Y., et al.: Radalign: Advancing radiology report generation with vision-language concept alignment. In: Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention. pp. 484–494. Springer (2025)
12. Guo, C., Pleiss, G., Sun, Y., et al.: On calibration of modern neural networks. In: Proceedings of the International Conference on Machine Learning. pp. 1321–1330. PMLR (2017)
13. Hamamci, I.E., Er, S., Menze, B.: Ct2rep: Automated radiology report generation for 3d medical imaging. In: International Conference on Medical Image Computing and Computer-Assisted Intervention. pp. 476–486. Springer (2024)
14. Holste, G., Wang, S., Jiang, Z., et al.: Long-tailed classification of thorax diseases on chest x-ray: A new benchmark study. In: Proceedings of the MICCAI Workshop on Data Augmentation, Labelling, and Imperfections. pp. 22–32. Springer (2022)
15. Hurst, A., Lerer, A., Goucher, A.P., et al.: Gpt-4o system card. arXiv preprint arXiv:2410.21276 (2024)
16. Irvin, J., Rajpurkar, P., Ko, M., et al.: Chexpert: A large chest radiograph dataset with uncertainty labels and expert comparison. In: Proceedings of the AAAI Conference on Artificial Intelligence. vol. 33, pp. 590–597 (2019)
17. Jin, H., Che, H., Lin, Y., et al.: Promptmrg: Diagnosis-driven prompts for medical report generation. In: Proceedings of the AAAI conference on artificial intelligence. vol. 38, pp. 2607–2615 (2024)

18. Johnson, A., Pollard, T., Mark, R., et al.: Mimic-cxr database. PhysioNet (2024)
19. Kullback, S.: Kullback-leibler divergence. Encyclopedia of Machine Learning pp. 581–583 (1951)
20. Li, W., Han, G., Wu, Y., et al.: Joint imbalance adaptation for radiology report generation. Journal of Healthcare Informatics Research pp. 1–23 (2025)
21. Liu, Z., Miao, Z., Zhan, X., Wang, J., Gong, B., Yu, S.X.: Large-scale long-tailed recognition in an open world. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 2537–2546 (2019)
22. Madhipati, R., Bhat, S., Buess, L., et al.: Cxr-cml: Improved zero-shot classification of long-tailed multi-label diseases in chest x-rays. In: Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention. pp. 119–129. Springer (2025)
23. Mahmood, R., Machado-Reyes, D., Wu, J., et al.: Phrase-grounded fact-checking for automatically generated chest x-ray reports. In: Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention. pp. 441–452. Springer (2025)
24. Ostmeier, S., Xu, J., Chen, Z., et al.: Green: Generative radiology report evaluation and error notation. In: Findings of the Empirical Methods in Natural Language Processing. pp. 374–390 (2024)
25. Papineni, K., Roukos, S., Ward, T., et al.: Bleu: a method for automatic evaluation of machine translation. In: Proceedings of the Association for Computational Linguistics. pp. 311–318 (2002)
26. Sellergren, A., Kazemzadeh, S., Jaroensri, T., Kiraly, A., Traverse, M., Kohlberger, T., Xu, S., Jamil, F., Hughes, C., Lau, C., et al.: Medgemma technical report. arXiv preprint arXiv:2507.05201 (2025)
27. Xiong, M., Hu, Z., Lu, X., et al.: Can llms express their uncertainty? an empirical evaluation of confidence elicitation in llms. In: Proceedings of the International Conference on Learning Representations (2024)
28. Xuan, W.M., Zeng, Q., Qi, H., et al.: Seeing is believing, but how much? a comprehensive analysis of verbalized calibration in vision-language models. In: Proceedings of the Empirical Methods in Natural Language Processing. pp. 1408–1450 (2025)
29. Zhang, S., Xu, Y., Usuyama, N., Xu, H., Bagga, J., Tinn, R., Preston, S., Rao, R., Wei, M., Valluri, N., et al.: Biomedclip: a multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs. arXiv preprint arXiv:2303.00915 (2023)
30. Zhang, T., Kishore, V., Wu, F., et al.: Bertscore: Evaluating text generation with bert. In: Proceedings of the International Conference on Learning Representations (2020)